

Instructions

TDS 500, TDS 600, TDS 700 & TLS 216 Floppy Disk Drive Emulator

Installation and Operation

TDS Floppy Emulator Firmware 1.0.0

TFE-0001-00

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Figures 2–2, 2–3 and 2–4 are reproduced from the Tektronix *TDS 500D, TDS 600C, TDS 700D & TDS 714L Service Manual*, 071-0627-02 (Figures 6–7, 6–8 and 6–21, pages 6–21, 6–22 and 6–45), copyright © Tektronix, Inc., under the Tektronix policy permitting free use of documentation for obsolete products. The installation procedures follow the structure of Tektronix instructions 061-4094-02, *TDS, TLS & CTS Families Floppy Disk Drive Replacement*.

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Service Safety Summary

WARNING. *The servicing instructions are for use by qualified personnel only. To avoid personal injury, do not perform any servicing unless you are qualified to do so. Refer to the General Safety Summary in the oscilloscope service manual before performing any service.*

Do Not Service Alone. Do not perform internal service on this product unless another person capable of rendering first aid and resuscitation is present.

Disconnect Power. To avoid electric shock, set the oscilloscope power switch to off and disconnect the power cord before removing the rear cover or cabinet.

Avoid Exposed Circuitry. To avoid injury, remove jewelry such as rings, watches, and other metallic objects. Do not touch exposed connections and components when power is present.

Use Care When Servicing With Power On. Dangerous voltages or currents may exist in this product. Disconnect power, remove battery (if applicable), and disconnect test leads before removing protective panels, soldering, or replacing components.

Prevent Electrostatic Discharge. The emulator and the oscilloscope circuit boards are sensitive to static discharge. Work only in a static-controlled environment and wear a grounded wrist strap.

Preface

These instructions describe how to install an SFR1M44-DU26 floppy disk drive emulator, running the TDS Floppy Emulator Firmware, in place of the 3.5 inch floppy disk drive of a TDS 500, TDS 600, TDS 700 or TLS 216 series instrument, and how to operate it.

The emulator presents a 1.44 Mbyte floppy disk to the oscilloscope. The contents of that disk are held in the emulator's internal memory and are exchanged with a USB flash drive by means of two buttons on the front of the emulator. The oscilloscope reads, writes and formats the disk exactly as it does a floppy disk in the original drive; no oscilloscope settings or firmware changes are required.

This document applies to firmware version 1.0.0 and above. Operation has been verified with a TDS 794D (oscilloscope firmware v8.0e). The other instruments listed in *Instruments* on page 1–1 use the same floppy disk drive and interface.

Related Manuals

The Tektronix instructions 061-4094-XX, *Floppy Disk Drive Replacement*, describe the replacement of the floppy disk drive with a new drive.

The service manual for the oscilloscope contains complete disassembly and reassembly information. The *TDS 500D*, *TDS 600C*, *TDS 700D* & *TDS 714L Service Manual* (071-0627-XX) is the source of Figures 2–2 to 2–4.

The user manual for the oscilloscope describes the save/recall and file utility functions used with the floppy disk.

Kit Description

Product Description

The kit replaces the floppy disk drive of the instrument with a floppy disk drive emulator. The emulator fits the drive bracket of the original drive, connects to the same 26-pin flexible cable, and takes its power from that cable; no other connection is made inside the instrument.

The emulator holds one 1.44 Mbyte disk in internal flash memory, which retains its contents when the instrument is powered off. A USB port on the front of the emulator accepts a USB flash drive. The buttons below the port load files from the flash drive onto the disk, save the files on the disk to the flash drive, and update the emulator firmware.

Instruments

Table 1–1: Instruments

Instrument	Serial numbers
TDS 500A/B/C/D	All serial numbers
TDS 600A/B/C	All serial numbers
TDS 700A/C/D/L	All serial numbers
TLS 216	All serial numbers

Kit Parts List

Table 1–2: Kit parts list

Quantity	Description
1 ea.	Floppy disk drive emulator: SFR1M44-DU26 (PCB SFRC2D.B), 26-pin 1 mm FFC interface, loaded with the TDS Floppy Emulator Firmware
1 ea.	Jumper link, 2.54 mm (fitted to the S1 position; see <i>Set the Jumper</i> on page 2–3)
1 ea.	Instructions (this document)
1 ea.	USB flash drive, FAT32 or exFAT (not supplied; any commercial flash drive; see <i>USB Flash Drive Requirements</i> on page 3–2)

Minimum Tools and Equipment List

Table 1–3: Minimum tools and equipment

Tool or equipment	Purpose
Torx driver with T-15 and T-20 heads	Drive bracket and rear cover
Phillips driver, small head	Drive mounting screw
Plastic or wooden tool, flat	Flexible-cable connector cam
Grounded wrist strap	Static protection
Torque driver, 16 inch-lbs	Rear cover screws
PC running Windows, Artery ISP Programmer, USB-A to USB-A cable	First installation of the firmware only (see <i>Load the Firmware</i> on page 2–1)

Installation Instructions

These instructions assume that you are familiar with the instrument. If additional assembly or disassembly details are required, refer to the service manual.



WARNING. To prevent electrical shock, ensure the mainframe power switch is in the off position before proceeding. Then, disconnect the instrument from the power source. Disassembly should only be attempted by qualified service personnel.



CAUTION. To prevent static-discharge damage, service the instrument only in a static-free environment. Observe standard handling precautions for static-sensitive devices while installing this kit. Always wear a grounded wrist and foot strap.

Load the Firmware

An emulator supplied with the TDS Floppy Emulator Firmware already installed does not require this procedure; go to *Set the Jumper* on page 2–3. An emulator supplied with its factory firmware must be loaded once, before it is installed in the instrument, through the USB bootloader built into its microcontroller. Later firmware versions are installed from a USB flash drive (see *Update the Firmware* on page 3–3).

Table 2–1: Load the firmware: requirements

Item	Requirement
Equipment required	PC running Windows; USB-A to USB-A cable; one jumper link (2.54 mm)
Software required	Artery ISP Programmer (Artery Technology), including its DFU driver installer
Firmware file	<code>tdsfloppy_install.hex</code> from the release folder of the firmware distribution (bootloader and firmware in one file)
Prerequisites	The emulator is not installed in an instrument and no USB flash drive is connected



CAUTION. *The factory firmware is erased permanently by this procedure. The factory firmware cannot be read out of the emulator and is not required for any later step.*

1. *Select the bootloader:* Fit a jumper link across pins 1 and 2 of header **J3** on the emulator circuit board (pin 1 is BOOT0, pin 2 is +3.3 V; see Figure 2–1).
2. *Install the DFU driver:* Run `Artery_DFU_DriverInstall.exe` from the `Artery_DFU_DriverInstall` folder of the Artery ISP Programmer package, and follow its prompts.
3. *Connect the emulator:* Connect the emulator USB port to the PC with the USB-A to USB-A cable. The emulator is powered from the cable and is detected by the PC as a DFU device.
4. *Select the device:* Start the Artery ISP Programmer. Set the port type to **USB DFU**, select the emulator in the device list, and press **Next** until the operation page appears.
5. *Remove the access protection:* Select **Enable/Disable Protection**; then select **DISABLE** and **Access Protection**, press **Next**, and confirm with **Yes**. The flash memory is erased. When the erase completes, press **Back**.
6. *Program the firmware:* Select **Download to device**, press **Add**, select `tdsfloppy_install.hex`, and press **Next**. The file is programmed and verified. Press **Ok**; then press **Close**.
7. *Restore normal start-up:* Disconnect the USB cable and remove the jumper link from header **J3**.

NOTE. *Use `tdsfloppy_install.hex`, not `tdsfloppy.hex`. The latter holds the firmware without its bootloader and does not start when programmed on its own.*

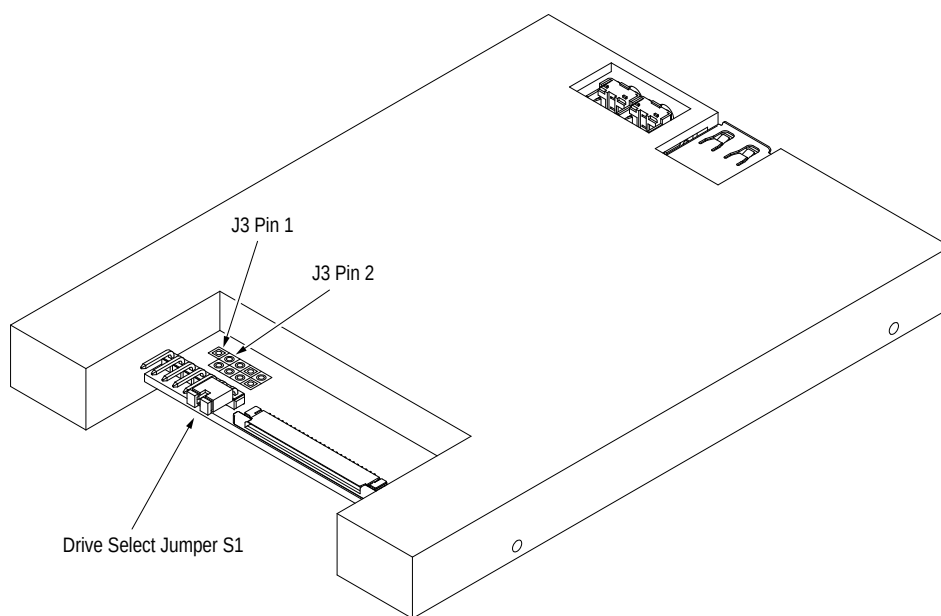


Figure 2–1: Emulator jumper locations: header J3 and the drive select jumper

Set the Jumper

The jumper block at the rear of the emulator, next to the 26-pin connector, sets the drive-select line and optional interface signals (see Figure 2–1). Table 2–2 lists the positions.

Table 2–2: Jumper positions

Position	Pins	Function	Setting
S1	Three-pin header, pins nearest the connector	Select by the drive-select line	Fitted
MO	Three-pin header, other two pins	Select by the motor-on line	Not fitted
JE	Two-pin header	720 kbyte (DD) drive (not supported)	Not fitted
JD	Two-pin header	Disk always reported installed	Not fitted

1. *Set the drive select:* Install one jumper link across the two pins closest to the 26-pin connector (position **S1**).
2. *Clear the other positions:* Remove any jumper link from positions **MO**, **JE** and **JD**.

Remove Rear Cover & Cabinet

Table 2-3: Remove rear cover and cabinet: requirements

Item	Requirement
Equipment required	One screwdriver with a size T-20 Torx tip

1. *Position the oscilloscope:* Install the front cover. Set the oscilloscope face down with the front cover on the work surface and the bottom facing up (see Figure 2-2).
2. *Disconnect power:* Unplug the power cord from its receptacle at the rear panel.
3. *Remove the rear cover:* Using the screwdriver with size T-20 Torx tip, remove the four screws securing the rear cover to the oscilloscope. Lift off the rear cover.
4. *Turn the oscilloscope over:* Set the oscilloscope so its top is down on the work surface and its front is facing you.
5. *Remove the cabinet:* Grasp the left and right edges of the cabinet at the rear. Pull upward to slide the cabinet off the oscilloscope. Do not bind or snag the cabinet on the oscilloscope internal cabling as you remove it.

NOTE. When reinstalling the four screws at the rear panel, use the torque driver to tighten the screws to 16 inch-lbs torque.

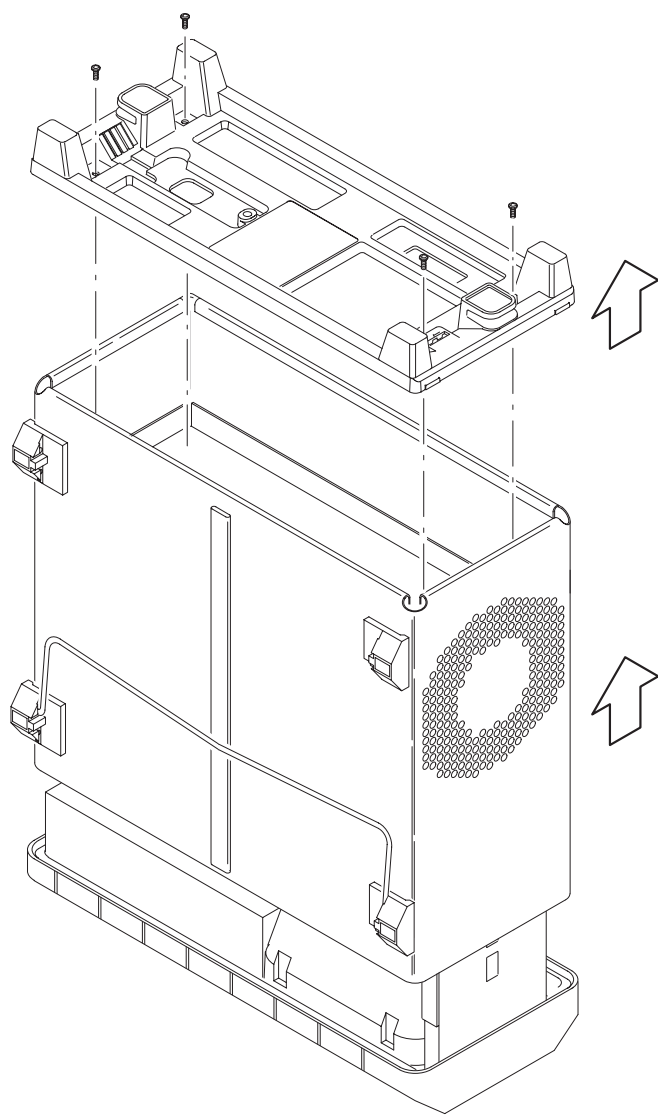


Figure 2-2: Rear cover and cabinet removal

Trim Ring Removal

Trim ring removal is optional. The drive assembly can be withdrawn toward the rear of the instrument with the trim ring in place, but removing the trim ring gives easier access to the drive and to the bezel opening when the emulator is aligned.

Table 2-4: Trim ring removal: requirements

Item	Requirement
Equipment required	None

1. *Remove the trim ring:* Grasp the trim ring by its top edge and pry it up and lift it forward to snap it off of the front subpanel (see Figure 2–3). When reinstalling, be sure the menu buttons are properly reinserted, align the trim ring to the front subpanel, and press it back on.

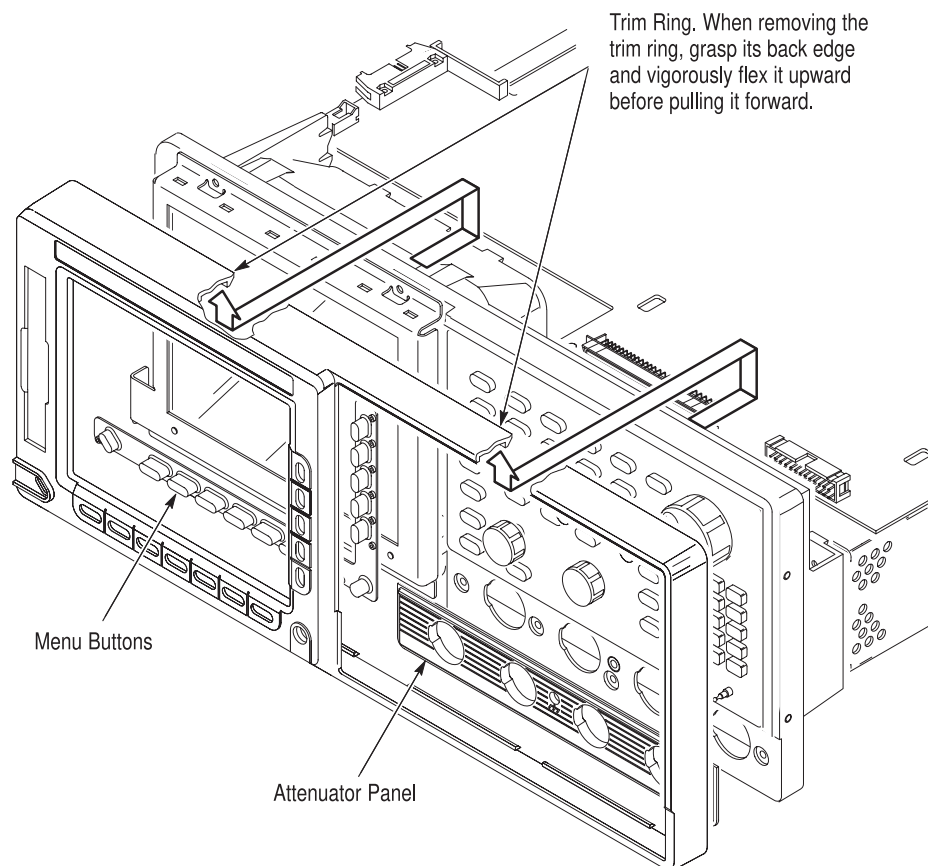


Figure 2–3: Front cover, trim ring, menu buttons, and attenuator panel removal (front cover not shown)

NOTE. Do not touch the carbon contact points on the menu buttons installed in the trim ring. Also, do not touch the contacts on the flex circuit exposed when you remove the trim ring.

Floppy Disk Drive Replacement

Table 2–5: Floppy disk drive replacement: requirements

Item	Requirement
Equipment required	One small head Phillips screwdriver
	One screwdriver with a size T-15 Torx tip
	One flat plastic or wooden tool
Optional items	3D-printed adapter bracket, if the instrument has none
	One M2.5 × 10 mm screw, for the printed adapter bracket
Prerequisites	The emulator is loaded with the firmware and its jumper is set (see page 2–1 and page 2–3)

1. *Remove the floppy disk drive* (see Figure 2–4):
 - a. *Release the drive bracket:* Remove the two T-15 Torx screws holding the drive bracket to the chassis. Set the screws aside for reassembly.
 - b. *Withdraw the drive:* Slide the bracket and floppy disk drive together toward the rear of the instrument. If the locking ears of the nearby connector interfere, disconnect that ribbon cable temporarily.
 - c. *Release the flexible cable:* Using the flat plastic or wooden tool, gently lift the cam of the drive connector that secures the flexible cable (some cams lift straight up, others swing up). Withdraw the cable.
 - d. *Separate the drive from the bracket:* Remove the Phillips screws that secure the drive and shields to the bracket. Keep the screws, and the adapter bracket and its M2.5 × 10 mm screw if fitted; discard or store the drive.

2. *Install the emulator* (see Figure 2–4):
 - a. *Orient the emulator*: Position the emulator in the bracket with its USB port above the two buttons, as seen from the front of the instrument.
 - b. *Connect the flexible cable*: Insert the flexible cable into the emulator connector as far as possible. Using the flat tool, gently press the connector cam into place to secure the cable. When inserted correctly, the side ears of the cam bottom out on the connector body. Verify that the cable and its contact fingers are straight.
 - c. *Secure the emulator*: Secure the emulator to the bracket with the two original M2.5 × 5 mm Phillips drive screws, and fit the adapter bracket with its M2.5 × 10 mm screw. Three screws hold the emulator.
 - d. *Install the bracket*: Install the bracket to the chassis using the two T-15 Torx screws removed in step 1. Reconnect any ribbon cable that was disconnected in step 1.

NOTE. *The emulator is narrower than the floppy disk drive, whose opening allows for an eject button. The emulator is held by the original FDD screws and by the front bezel of the instrument. Depending on the floppy disk drive fitted at the factory, the instrument has an adapter bracket (the alternate mounting bracket in Figure 2–4) and an M2.5 × 10 mm screw, or neither. If no adapter bracket is present, one can be 3D printed from the supplied files (STL and STEP formats); the printed bracket requires one additional M2.5 × 10 mm screw.*

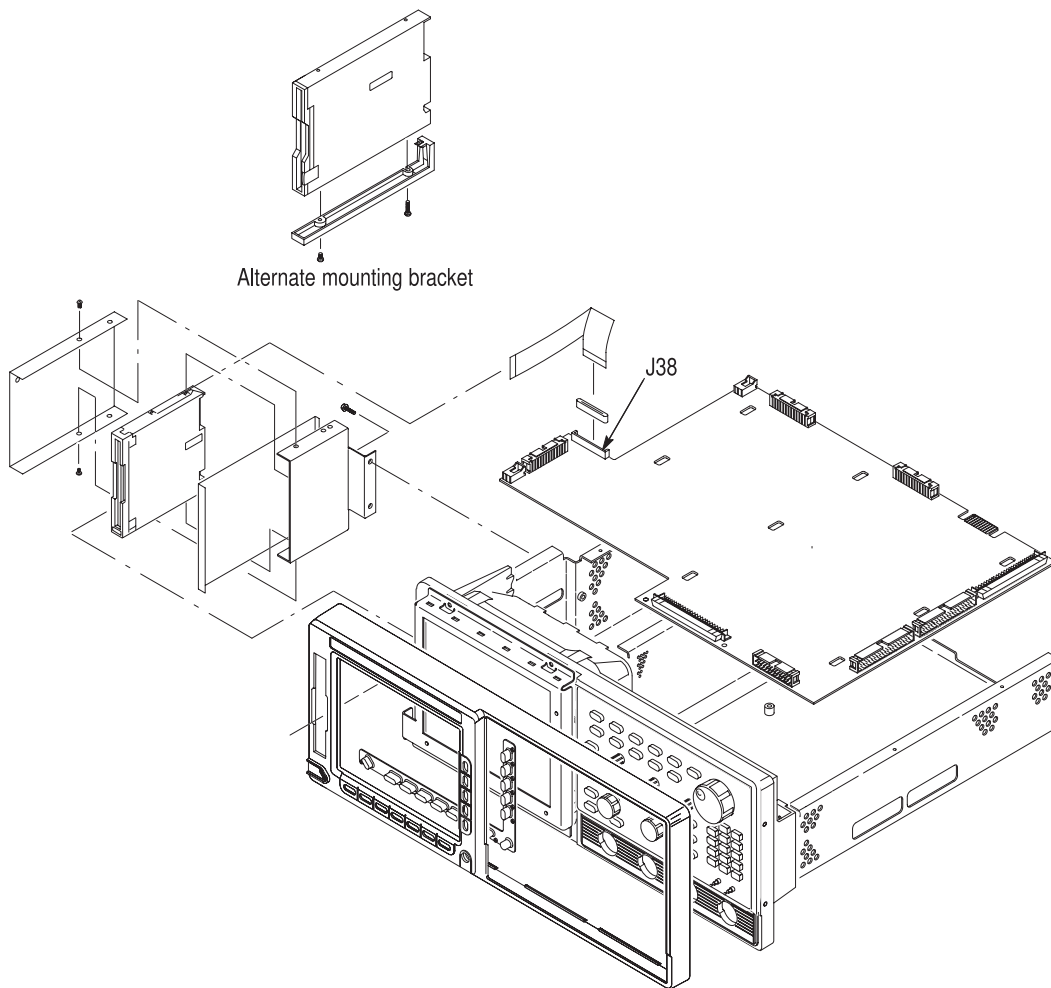


Figure 2-4: Floppy disk drive removal

Reassembly



CAUTION. On some models, large heat sinks protrude inside the instrument. Do not disturb them when you slide the cabinet back into place.

1. Reinstall the trim ring (if it was removed) by performing the *Trim Ring Removal* section in reverse order.
2. Reinstall the rear cover & cabinet by performing the *Remove Rear Cover & Cabinet* section in reverse order. Tighten the four rear panel screws to 16 inch-lbs torque.

Verify the Installation

Table 2-6: Verify the installation: requirements

Item	Requirement
Equipment required	None
Prerequisites	The instrument is reassembled

1. *Power on the oscilloscope:* Connect the power cord and set the power switch to on. Wait for the oscilloscope to complete its power-on sequence.
2. *Access the floppy disk:* Display the directory of the floppy disk (`fd0:`) with the oscilloscope file utilities, as described in the oscilloscope user manual.
3. *Verify that the emulator is operational:* Confirm that the following statements are true.
 - The green indicator (if fitted) lights while the oscilloscope accesses the disk.
 - The oscilloscope displays the directory of the disk without an error message. A new emulator presents an empty, formatted disk.
 - The red indicator is off when no transfer is in progress.
4. *Verify saving:* Save the current setup to the floppy disk from the oscilloscope save/recall menus, or save a screenshot to the floppy disk with the oscilloscope hardcopy function. Confirm that the file appears in the directory of `fd0:`.

Operating Instructions

Controls and Indicators

The emulator is mounted with its USB port above its two buttons. The upper button is the button nearest the USB port.

Table 3–1: Controls and indicators

Control or indicator	Function
USB port	Accepts a USB flash drive. Nothing is transferred when a flash drive is inserted; transfers start only from the buttons.
Upper button (DATA OUT)	Press and release: save every file on the internal disk to the flash drive, verify them, then erase the internal disk.
Lower button (DATA IN)	Press and release: replace the internal disk with the next group of files from the flash drive.
Both buttons	Hold both for 3 seconds, then release: update the emulator firmware from the flash drive.
Red indicator	Transfer and result indications (see Table 3–4).
Green indicator (if fitted)	Lights while the oscilloscope selects the drive (disk access in progress). Not fitted on some emulators.

NOTE. A button action takes effect when the button is released. Pressing both buttons briefly does nothing, and starting a two-button hold never starts a single-button transfer.

The Internal Disk

The internal disk is a 1.44 Mbyte MS-DOS (FAT12) disk: 80 cylinders, 2 heads, 18 sectors of 512 bytes. The oscilloscope reads, writes and formats it as a floppy disk. Its contents are held in flash memory and are retained when the instrument is powered off.



CAUTION. Do not power off the instrument while the oscilloscope is saving a file. As with a floppy disk, a save interrupted by power loss can leave the disk directory incomplete.

USB Flash Drive Requirements

Table 3–2: USB flash drive requirements

Characteristic	Requirement
File system	FAT12, FAT16, FAT32 or exFAT
Partitioning	MBR, GPT, or unpartitioned
Logical sector size	512 bytes
Capacity	Up to 2 Tbyte
Files used	Files in the root directory only
Not supported	NTFS and other file systems; flash drives with other sector sizes (indicated by three slow blinks; reformat the flash drive as exFAT or FAT32)

The oscilloscope uses MS-DOS 8.3 file names. A file on the flash drive whose name does not fit that form appears on the internal disk under a short name of the form NAME~1.EXT (for example, Scope capture 1.bmp appears as SCOPEC~1.BMP).

Loading Files from a USB Flash Drive

DATA IN copies the files in the root directory of the flash drive, in alphabetical order, onto the internal disk, until the next file does not fit. Each further press loads the next group; after the last group, the next press starts again from the first. Removing the flash drive restarts the sequence from the first group.

Directories, hidden and system files, files whose names start with a period, files larger than 1.44 Mbyte, and the update file UPDATE.UPD are not copied.



CAUTION. DATA IN replaces the entire contents of the internal disk. Files saved by the oscilloscope that are not yet on the flash drive are lost. Save them first with DATA OUT.

1. *Insert the flash drive:* Insert the flash drive into the emulator USB port.
2. *Load the files:* Press and release the lower button (DATA IN). The red indicator flashes while the files are copied.
3. *Check the result:* CHECK that the red indicator goes off without a blink pattern (see Table 3–4).
4. *Use the files:* Access the files on fd0: from the oscilloscope. The oscilloscope detects the disk change.

Saving Files to a USB Flash Drive

DATA OUT copies every file on the internal disk to the flash drive, keeping the date and time of each file. Directories created by the oscilloscope are recreated on the flash drive with their contents. Each copy is read back from the flash drive and compared with the original. Only when every file has been copied and verified is the internal disk erased, ready for new data.

Files on the flash drive are never overwritten. A file whose name already exists on the flash drive is saved as NAME_1.EXT, NAME_2.EXT, and so on; an identical copy already on the flash drive is not copied again. If any file cannot be copied or does not verify, the internal disk is left unchanged and no partial copy remains on the flash drive.

1. *Insert the flash drive:* Insert the flash drive into the emulator USB port.
2. *Save the files:* Press and release the upper button (DATA OUT). The red indicator flashes while the files are copied and verified.
3. *Check the result:* CHECK that the red indicator goes off without a blink pattern (see Table 3–4). The internal disk is now empty.

NOTE. A transfer requested while the oscilloscope is using the drive starts when the oscilloscope has left the drive idle for 1.5 seconds. The oscilloscope keeps the drive selected for several seconds after a save. If the drive is still in use after 15 seconds, the transfer is not performed and the error pattern is shown; press the button again.

Formatting the Internal Disk

The internal disk is formatted from the oscilloscope file utilities, in the same way as a floppy disk. A format takes approximately 3 minutes. DATA OUT also leaves the internal disk empty after a successful save.

Update the Firmware

New firmware is supplied as the file UPDATE . UPD. The file is checked completely before anything is changed; a file that is damaged, incomplete or made for other hardware is refused. If power is lost during the update, the update completes at the next power-on. The internal disk is not affected.

Table 3–3: Update the firmware: requirements

Item	Requirement
Equipment required	USB flash drive with UPDATE . UPD in its root directory
Time required	Approximately 5 seconds

1. *Insert the flash drive:* Insert the flash drive into the emulator USB port.
2. *Arm the update:* Press and hold both buttons. After 3 seconds the red indicator lights steadily.
3. *Start the update:* Release both buttons. The red indicator flashes while the file is checked and copied; the emulator then restarts and installs the firmware.
4. *Check the result:* CHECK that the emulator resumes normal operation within 10 seconds. Two quick blinks indicate that no UPDATE .UPD was found; six quick blinks indicate that the file was refused. In both cases the installed firmware is unchanged.

Indicator Patterns

Table 3–4: Red indicator patterns

Pattern	Meaning
Off	Idle
Flashing, 4 times per second	Transfer in progress; do not remove the flash drive
On steadily	Both buttons held for 3 seconds; release to update the firmware
2 quick blinks	Nothing to load (no suitable files, or no UPDATE.UPD); internal disk unchanged
3 slow blinks	Flash drive format not supported; reformat as exFAT or FAT32; internal disk unchanged
1 long blink (1 s)	Transfer cancelled (see <i>Cancelling a Transfer</i>)
6 quick blinks	Error: no flash drive, USB error, verification failure, update file refused, or drive in use for more than 15 seconds; internal disk unchanged
Fast blinking, continuous	Internal flash memory failed. At power-on the emulator then presents no disk and refuses transfers
Fast double blink, repeating	No valid firmware installed; load the firmware (see page 2–1)

Cancelling a Transfer

Press the button that started a transfer again while the red indicator flashes; the indicator then gives one long blink. A cancelled DATA OUT deletes any partly copied file and leaves the internal disk unchanged; files already copied remain. A cancelled DATA IN keeps the files loaded so far, and the next press loads the same group again; the previous contents cannot be restored. A firmware update cannot be cancelled.

Troubleshooting

Table 4–1: Troubleshooting

Symptom	Possible cause	Action
Oscilloscope reports no disk or a drive error; green indicator (if fitted) never lights	Flexible cable not seated, or jumper not at S1	Reseat the cable in the connector cam; check the jumper (page 2–3)
Directory shows unexpected files	A DATA IN replaced the disk	Load the required group with DATA IN, or save with DATA OUT
3 slow blinks after a button press	Flash drive is NTFS, unformatted, or not 512-byte sectors	Reformat the flash drive as exFAT or FAT32
2 quick blinks after DATA IN	No suitable files in the root directory	Copy the files to the root directory of the flash drive
6 quick blinks after a button press	No flash drive, USB error, or the oscilloscope was using the drive	Reinsert the flash drive, wait for the oscilloscope to finish, and press again
A file is missing after DATA IN	File larger than 1.44 Mbyte, hidden, in a directory, or in a later group	Press DATA IN again for the next group; move the file to the root directory
Copies named NAME_1.EXT on the flash drive	A file of that name already existed	Normal operation; existing files are never overwritten
Fast double blink at power-on	No valid firmware	Load the firmware (page 2–1)
Fast continuous blinking, also after power-on	Internal flash memory failing (the indication is kept until the disk is erased)	Save the files with DATA OUT; if it recurs, replace the emulator

Specifications

Table 5–1: Floppy disk interface

Characteristic	Description
Emulated drive	3.5 inch, 1.44 Mbyte (2HD), 300 rpm, 500 kbit/s MFM
Disk format	80 cylinders, 2 heads, 18 sectors of 512 bytes per track; the head steps to cylinder 80
Interface	26-pin slim floppy interface, 1 mm FFC with cam-lock connector; drive select 1 (jumper S1)
Power	+5 V from the 26-pin interface cable
Ready	READY 480 ms after drive select, as a drive after motor spin-up
Write storage	Starts 0.1 s after each write ends, through a power-safe journal: a power loss affects only the write in progress. The head does not wait for flash memory

Table 5–2: Storage and USB

Characteristic	Description
Internal disk	1.44 Mbyte in SPI flash memory, retained without power
USB	USB 2.0 full-speed host, mass-storage class
Flash drive file systems	FAT12, FAT16, FAT32, exFAT; MBR, GPT or unpartitioned; 512-byte sectors; up to 2 Tbyte
Load time (DATA IN)	Approximately 6 s for 470 kbyte
Save time (DATA OUT)	Approximately 11 s for 470 kbyte, including verification
Firmware update	UPDATE.UPD, header-checked and CRC-verified; approximately 5 s; protected against power loss